

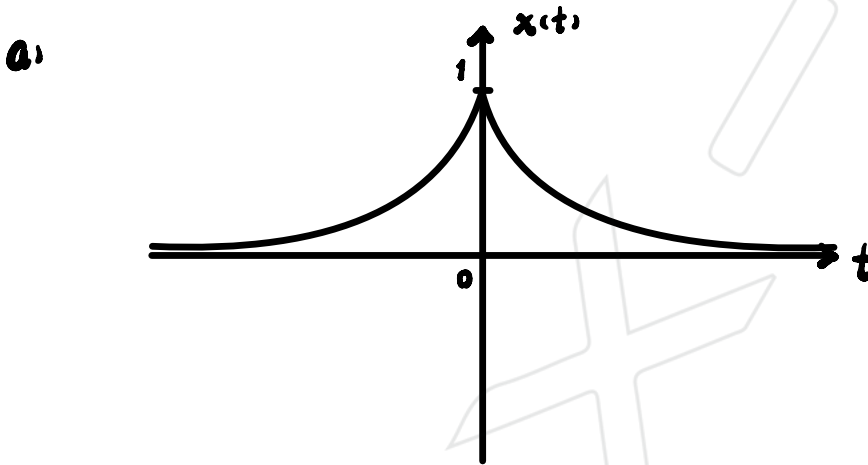
Mechatronics Tutorial Sheet 1: Fourier Series & Fourier Transform

1. Consider the CT signal

$$x(t) = e^{-a|t|},$$

where $a \in \mathbb{R}$ is a constant parameter.

- Suppose $a > 0$. Sketch the signal $x(t)$.
- Suppose $a > 0$. Compute the Fourier transform of $x(t)$.
- Suppose now that $a < 0$. Is it still possible to compute the Fourier transform of the signal $x(t)$ in this case?



$$b) \quad x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{x}(j\omega) e^{j\omega t} d\omega \quad \text{where } \omega = \frac{2\pi k}{T}$$

$$\therefore \hat{x}(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} e^{-a|t|} e^{-j\omega t} dt$$

$$= \int_{-\infty}^0 e^{(a-j\omega)t} dt + \int_0^{\infty} e^{-(a+j\omega)t} dt$$

$$\begin{aligned}
 \therefore \hat{X}(j\omega) &= \frac{1}{a-j\omega} [e^{(a-j\omega)t}]_{-\infty}^0 + \frac{1}{-a-j\omega} [e^{-(a+j\omega)t}]_0^{\infty} \quad (a > 0) \\
 &= \frac{1}{a-j\omega} (1-0) + \frac{1}{-a-j\omega} (0-1) \\
 &= \frac{1}{a-j\omega} + \frac{1}{a+j\omega} \\
 &= \frac{2a}{a^2 + \omega^2}
 \end{aligned}$$

C) No, because the $x(t)$ becomes unbounded with bounded input t , not fulfill the BIBO stable condition

2. Determine the Fourier series coefficients for each of the following discrete-time periodic signals:

a) $x[n] = \sin\left(\frac{\pi n}{3}\right) \cos\left(\frac{\pi n}{2}\right)$

b)

$$x[n] = \begin{cases} 2 & \text{if } |n| \leq 1 \\ -1 & \text{if } |n| = 2 \\ 0 & \text{if } |n| = 3 \end{cases}$$

where $x[n]$ has period 6.

a) $C_k = \frac{1}{N} \sum_{n=-N}^N x[n] e^{-jk\omega_0 n}$ where $\omega_0 = \frac{2\pi}{N}$ and $N = 12$

$$= \frac{1}{12} \sum_{n=-6}^6 \sin\left(\frac{\pi n}{3}\right) \cos\left(\frac{\pi n}{2}\right) e^{-j\frac{k\pi n}{6}}$$

$$= \frac{1}{12} (x[-6] e^{-jk\pi} + x[6] e^{jk\pi} + \dots + x[0])$$

$$= \frac{1}{12} [0 + 0 + 0 + 0 + \sin\left(\frac{4\pi}{3}\right) \cos(2\pi) (e^{-\frac{2}{3}jk\pi} - e^{\frac{2}{3}jk\pi}) + 0 + 0 + \sin\left(\frac{2\pi}{3}\right) \cos(\pi) (e^{-\frac{1}{3}jk\pi} - e^{\frac{1}{3}jk\pi}) + 0 + 0 + 0]$$

$$= -\frac{1}{6} j \left[\sin\left(\frac{2\pi}{3}\right) \cos(\pi) \sin\left(\frac{1}{3}k\pi\right) \right]$$

$$= \frac{\sqrt{3}}{12} j \sin\left(\frac{1}{3}k\pi\right)$$

$$\therefore C_k = \begin{cases} 0 & k \neq \pm 1 + 2n, n \in \mathbb{R} \\ \frac{1}{4}j & k = 1 + 2n, n \in \mathbb{R} \\ -\frac{1}{4}j & k = -1 + 2n, n \in \mathbb{R} \end{cases}$$

2. Determine the Fourier series coefficients for each of the following discrete-time periodic signals:

a) $x[n] = \sin\left(\frac{\pi n}{3}\right) \cos\left(\frac{\pi n}{2}\right)$

b)

$$x[n] = \begin{cases} 2 & \text{if } |n| \leq 1 \\ -1 & \text{if } |n| = 2 \\ 0 & \text{if } |n| = 3 \end{cases}$$

where $x[n]$ has period 6.

$$b) \quad C_k = \frac{1}{N} \sum_{n=-\infty}^{\infty} x[n] e^{-jk\omega_0 n} \quad \text{where } \omega_0 = \frac{2\pi}{N} \text{ and } N = 6$$

$$= \frac{1}{N} \sum_{n=-3}^3 x[n] e^{-j\frac{k\pi n}{3}}$$

$$= \frac{1}{6} (x[0] e^0 + x[1] e^{-j\frac{k\pi}{3}} + x[-1] e^{j\frac{k\pi}{3}} + \dots)$$

$$= \frac{1}{6} (2 + 2[2\cos(\frac{k\pi}{3})] - 1[2\cos(\frac{2k\pi}{3})] + 0)$$

$$= \frac{1}{3} + \frac{2}{3}\cos(\frac{k\pi}{3}) - \frac{1}{3}\cos(\frac{2k\pi}{3})$$

3. Given the Fourier series coefficients

$$c_k = \cos\left(\frac{k\pi}{4}\right) + j \sin\left(\frac{k\pi}{2}\right),$$

determine the corresponding discrete-time periodic signal.

$$x[n] = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 n} \quad \text{where } \omega_0 = \frac{2\pi}{N} \text{ and } N = 8$$

$$\begin{aligned} \therefore x[n] &= \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 n} = c_0 + c_1 e^{j\frac{\pi n}{4}} + c_{-1} e^{-j\frac{\pi n}{4}} + \dots + c_{-4} e^{-jn\pi} \\ &= 1 + 2[\cos(\frac{\pi}{4})](e^{j\frac{\pi n}{4}} + e^{-j\frac{\pi n}{4}}) + 2[\sin(\frac{\pi}{2})](e^{j\frac{\pi n}{4}} - e^{-j\frac{\pi n}{4}}) + \dots \\ &= 1 + 4\cos(\frac{\pi}{4})\cos(\frac{\pi n}{4}) - 4\sin(\frac{\pi}{2})j\sin(\frac{\pi n}{4}) + \dots \\ &= 1 + 2\sqrt{2}\cos(\frac{\pi n}{4}) - 2\sqrt{2}\cos(\frac{3\pi n}{4}) - 4\cos(n\pi) \\ &\quad - j[4\sin(\frac{\pi n}{4}) - 4\sin(\frac{3\pi n}{4})] \\ &= 1 + 4\sqrt{2}\cos(\frac{\pi n}{4}) - 4\cos(n\pi) \end{aligned}$$

$$x[n] = \sum_{k=-3}^4 \left[\frac{1}{2}(e^{j\frac{k\pi}{4}} + e^{-j\frac{k\pi}{4}}) + \frac{1}{2j}(e^{j\frac{k\pi}{4}} - e^{-j\frac{k\pi}{4}}) \right] e^{j\frac{k\pi n}{4}}$$

$$= \begin{cases} 4 & \text{if } n = \pm 1 \\ -4j & \text{if } n = -2 \\ 4j & \text{if } n = 2 \\ 0 & \text{otherwise} \end{cases}$$

4. Compute the Discrete-time Fourier transform of the function

$$x[n] = \cos(\omega_0 n).$$

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} \hat{x}(e^{j\omega}) e^{j\omega n} d\omega \quad \text{where } \omega = \frac{2k\pi}{N}$$

$$\therefore \hat{x}(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$= \sum_{n=-\infty}^{\infty} \frac{1}{2} (e^{j\omega_0 n} + e^{-j\omega_0 n}) e^{-j\omega n}$$

$$= \sum_{n=-\infty}^{\infty} \frac{1}{2} (e^{j(\omega_0 - \omega)n} + e^{-j(\omega_0 - \omega)n}) \cos(k\omega_0 n) - j \sin(k\omega_0 n)$$

$$= \sum_{n=-\infty}^{\infty} \cos(\omega_0 n) \cos(k\omega_0 n) - j \sum_{n=-\infty}^{\infty} \cos(\omega_0 n) \sin(k\omega_0 n)$$

$$= 2 \sum_{n=0}^{\infty} \cos(\omega_0 n) \cos(k\omega_0 n) - 0$$

$$= 2 \sum_{n=0}^{\infty} \cos(\omega_0 n) \cos(k\omega_0 n)$$

$$x[n] = \cos(\omega_0 n) = \frac{1}{2} (e^{j\omega_0 n} + e^{-j\omega_0 n}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \hat{x}(e^{j\omega}) e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} \hat{x}(e^{j(\omega + 2\pi k)}) e^{j\omega n} d\omega$$

$$\therefore \hat{x}(e^{j\omega}) = \pi \sum_{k=-\infty}^{\infty} [\delta(\omega - \omega_0 - 2\pi k) + \delta(\omega + \omega_0 - 2\pi k)]$$

5. A stable DT LTI system has transfer function

$$\hat{H}(e^{j\omega n}) = 1.$$

If the input of the system is a dirac impulse, what will the output be?

$$If \ x[n] = \delta[n] = \begin{cases} 1 & \text{for } n=0 \\ 0 & \text{otherwise} \end{cases}$$

$$\hat{X}(e^{j\omega}) = \sum_{n=-\infty}^{\infty} \delta[n] e^{-j\omega n} = 1 = e^0 \rightarrow x[n] = \delta[n]$$

$\therefore$ for $y[n] = x[n] * h[n]$,

$$\hat{Y}(e^{j\omega}) = \hat{X}(e^{j\omega}) \cdot \hat{H}(e^{j\omega}) = 1$$

$$\therefore y[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} \hat{Y}(e^{j\omega}) e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j\omega n} d\omega$$

$$= -\frac{j}{2\pi n} [e^{j\omega n}]_{-\pi}^{\pi}$$

$$= -\frac{j}{2\pi n} (e^{jn\pi} - e^{-jn\pi})$$

$$= -\frac{j}{2\pi n} [2j \sin(n\pi)]$$

$$= \frac{\sin(n\pi)}{\pi n}$$